Additional exercises for queueing theory
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Problem 1. Prove that the mean number of jobs waiting or in service for a M/D/1 queue with mean arrival rate λ and mean service rate μ is always lower than in an M/M/1 queue with the same parameters.

Problem 2. Suppose that the service time distribution of a queue whose utilization is 80% is described by $C_S^2 = 3$. Compare the mean time in the system for this queue with that of a M/M/1 queue with the same mean arrival and departure rates.

Problem 3. By instrumenting the time when a job starts service in the MacroHard Doors operating system, you discover that, for the first 1000 submitted jobs, the sum of service times is 412 seconds, and the sum of the squared service times is 1425 seconds^2. Assuming that jobs arrive to the system as a Poisson process and the mean utilization is 0.75, what is the mean number of jobs in the system, including the one in service?